The Uncovered Block Frontier in the TPC-17/18 Route

Eligibility Partitions and a Coefficientwise All-Block Obstruction

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Abstract

TPC-134 gives a complete dyadic prefix–tail archive, whereas the analytic theorems of TPC-17 and TPC-18 apply only on a strict block domain. We make this mismatch exact. A canonical maximal-integer policy selects $D_0(b)$ from the published Maynard inequalities, the finite-row leakage condition, and the TPC-18 tail geometry. The actual blocks partition into eligible and frontier sets, giving

$$B_{h_0,\delta}(X) = S_{\mathcal{E}}^{\text{prefix}} + S_{\mathcal{E}}^{\text{tail}} + S_{\mathcal{F}}$$

coefficientwise. The published TPC-17 theorem makes only the first term soft. For every nonzero W and all sufficiently large X, there is a native column with $\ell \asymp U = \sqrt{R}$; every dyadic child of this column has $L \le 2R$ and therefore lies in the frontier. Hence the eligible-only selector has a nonzero formal column defect in the path matrix. This stops only the declared eligible-only compiler. It proves neither that $S_{\mathcal{F}}$ is large nor that a different route is impossible. The partition is L0, the scoped obstruction is L1, and there is no new positive L2 estimate or prime-pair theorem.

1 One canonical eligibility policy

Retain the notation

$$R = \lfloor X^{1/2-\delta} \rfloor, \qquad V = \lfloor R^{1/2} \rfloor$$

and the actual dyadic blocks

$$b = (L, K), \qquad L = 2^{j_L}, \quad K = 2^{j_K}$$

from TPC-134 [3]. Fix once and for all a rational margin $\eta > 0$. We use only the formally published Maynard layer [2, 4]; the version-locked Li extension is a different route and is not silently mixed into this one.

Definition 1.1 (Admissible integer cutoff)**.** An integer $D \ge 1$ is admissible for b when

$$L > 2R, \qquad K > 2V, \qquad D < R, \qquad 2D < V, \tag{1}$$

$$LDR \le X^{1-\eta}, \qquad DL^2 \le X^{1-\eta}, \tag{2}$$

$$D^{12}L^7 \le X^{4-\eta}, \qquad D^{20}L^{19} \le X^{10-\eta}. \tag{3}$$

Define $D_0(b)$ to be the largest admissible integer, and set $D_0(b) = 0$ when none exists.

The maximum exists because $2D < V$ gives a finite candidate set. Every comparison is a rational-power inequality. Clearing denominators gives an exact integer test; no floating tolerance is part of the policy.

Definition 1.2 (Eligible and frontier blocks)**.** A block is *eligible* if $D_0(b) > 0$, and otherwise belongs to the frontier. Thus

$$\mathcal{B}_X = \mathcal{E}_X \sqcup \mathcal{F}_X.$$

For auditing, frontier blocks receive the first applicable reason:

$$L \le 2R, \quad K \le 2V, \quad \text{no integer with } 2D < V, \quad \text{or published-profile failure.}$$

2 Exact route partition

Let S_b^{prefix} and S_b^{tail} be the TPC-134 terminal sums at the canonical $D_0(b)$.

Theorem 2.1 (Eligible/frontier decomposition)**.** *For every X,*

$$\boxed{B_{h_0,\delta}(X) = S_{\mathcal{E}}^{\text{prefix}} + S_{\mathcal{E}}^{\text{tail}} + S_{\mathcal{F}},}$$

where

$$S_{\mathcal{E}}^{\text{prefix}} = \sum_{b\in\mathcal{E}_X} S_b^{\text{prefix}}, \quad S_{\mathcal{E}}^{\text{tail}} = \sum_{b\in\mathcal{E}_X} S_b^{\text{tail}}, \quad S_{\mathcal{F}} = \sum_{b\in\mathcal{F}_X} (S_b^{\text{prefix}} + S_b^{\text{tail}}).$$

This is a disjoint row partition of the complete TPC-134 path matrix.

Proof. TPC-134 gives the exact sum over every block. Partition its block indices by the deterministic predicate in definition 1.1. ☐

Corollary 2.2 (What the published prefix theorem supplies)**.** *On the fixed published-core scope above, TPC-17 gives, for every $A > 0$,*

$$S_{\mathcal{E}}^{\text{prefix}} = O_A(X(\log X)^{-A}).$$

Consequently

$$B_{h_0,\delta}(X) = S_{\mathcal{E}}^{\text{tail}} + S_{\mathcal{F}} + O_A(X(\log X)^{-A}).$$

Proof. Every eligible block satisfies the hypotheses of the cumulative prefix theorem by construction [4]. The dyadic support gives $O_W(\log X)$ relevant block pairs, so arbitrary logarithmic saving survives the sum. No estimate is applied to $\mathcal{F}_X$. ☐

3 A low-source frontier column

The following result is coefficientwise. It does not require the numeric coefficient of its witness to be nonzero.

Theorem 3.1 (Scoped all-block obstruction)**.** *Assume $W \not\equiv 0$. For all sufficiently large X, the TPC-134 archive contains a support-envelope native column all of whose dyadic children lie in $\mathcal{F}_X$. The corresponding formal column of the path matrix is present regardless of whether its arithmetic coefficient later evaluates to zero. Therefore, as operators on native columns,*

$$\boxed{P_{\mathcal{E}} M_{134} \ne M_{134}.}$$

In particular, the eligible TPC-17/18 block family cannot by itself be a coefficientwise all-block archive.

Proof. There is a closed interval $[c_1, c_2] \Subset (0, \infty)$ on which W is nonzero. By Bertrand's postulate, for large X choose a prime

$$U < \ell \leq 2U$$

[1]. The interval

$$[c_1X/\ell, c_2X/\ell]$$

has length tending to infinity, so it contains an integer $k > V$. Take the opened divisor $d = 1$. Then $W(\ell k/X) \neq 0$, hence this is a weight-active TPC-15 support-envelope record. We do not infer that its full coefficient is nonzero: the factor $r_Q(\ell k + h_0)$ may still vanish.

If a dyadic child has nonzero ℓ-weight, the support of ψ gives

$$\ell/2 < L < 2\ell \leq 4U.$$

Since $U = \lfloor\sqrt{R}\rfloor$, one has $4U \leq 2R$ for all sufficiently large R. Every child therefore fails $L > 2R$ and belongs to $\mathcal{F}_X$. The full column sum is one by TPC-134, while its eligible part is zero. □

Remark 3.2 (Precisely what is stopped). Theorem 3.1 stops a claimed derivation of exact coefficientwise cover using only eligible blocks. It does not rule out

$$S_{\mathcal{F}} = o(X),$$

a separate exact frontier transform, the asymmetric Vaughan route, or a scalar identity with a theorem-backed soft term. TPC-18 itself states its global all-block relation as an additional hypothesis [5]. We do not turn that warning into a global impossibility result.

4 Machine certificate and decision rules

The standard-library program `experiments/tpc135_domain_cover_audit.py` reads every committed TPC-134 sample path, recomputes the maximal integer cutoff, assigns one exhaustive reason to each block, and finds a native column whose children all have reason `L_TOO_LOW`. A separate large-integer regression exercises a genuinely eligible parameter cell. The sample witness is explicitly typed as a formal support-envelope column with undecided numerical coefficient. The program rejects a wrong reason, a missing block, a false eligible witness sum, a false nonzero-coefficient promotion, and any manifest that asserts a frontier scalar bound. Its `-check` mode never writes.

Statement	Status
Eligible/frontier set partition	Exact L0.
Low-source coefficientwise witness	Proved, scoped L1 obstruction.
Published eligible-prefix estimate	Imported restricted arithmetic theorem; not new here.
Frontier scalar $o(X)$ or lower bound	Neither is proved.
New growing fixed-h_0 power saving	None; no new positive L2.

Verdicts. A claim that eligible blocks alone are the exact coefficientwise archive receives STOP-DECLARED-ROUTE. The absence of a frontier theorem or map receives NOT-TESTABLE. If a future theorem supplies either, this scoped stop is discharged without changing the theorem proved here.

Explicit exclusions. No statement proves that $S_{\mathcal{F}}$ is large, that the hard packet is not $o(X)$, that every architecture fails, or that parity cannot be breached. There is no Hardy–Littlewood asymptotic, prime-pair lower bound, or twin-prime theorem.

References

[1] Tom M. Apostol. *Introduction to Analytic Number Theory.* Springer, 1976.

[2] James Maynard. *Primes in Arithmetic Progressions to Large Moduli I: Fixed Residue Classes*, volume 306 of *Memoirs of the American Mathematical Society.* American Mathematical Society, 2025. Number 1542.

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